

# Non-Nested Sampling Encoders: A Literal Counterexample and a Repair

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## Source and question

The source is Janek Goedeke and Pascal Fernsel, *New universal operator approximation theorem for encoder-decoder architectures (Preprint)*, arXiv:2503.24092.

After Theorem 3.20, the authors ask whether the conclusion remains true when the sampling points are not nested:

$$\{y_1^{(n)}, \dots, y_{k(n)}^{(n)}\} \quad \text{chosen separately for each } n.$$

The relevant source passage is shown in Figure 1.

$$f = \sum_{i=1} c_i(f) b_i = \sum_{i=1} f(y_i) b_i$$

holds for every  $f \in \mathcal{C}(\Omega, \mathbb{K})$ . However, such a Schauder basis representation, in which the unique coefficient functionals  $c_i$  sample the function  $f$ , is impossible, as shown in Theorem B.2 in the appendix.  $\square$

In the theorem above, the sampling points for  $E_{n+1}$  were a refinement of the sampling points for  $E_n$ . If, more generally, individual sampling points  $\{y_1^{(n)}, \dots, y_{k(n)}^{(n)}\}$  are chosen for each  $n \in \mathbb{N}$ , it remains an open question whether the previous theorem still holds. Let us close the discussion about Theorem 3.20 with a final remark on the assumption of  $k(n)$  to be unbounded. If  $\{y_1^{(n)}, \dots, y_{k(n)}^{(n)}\}$  is an  $\frac{1}{n}$ -covering of a set  $\Omega \subseteq \mathbb{K}^d$  having a non-zero Lebesgue measure  $\text{Vol}(\Omega) > 0$ , the unboundedness of  $k(n)$  follows from the translation-invariance and  $\sigma$ -subadditivity of the Lebesgue measure: If  $k(n)$  had an upper bound  $s < \infty$ , it would follow that

$$\text{Vol}(\Omega) \leq \sum_{i=1}^{k(n)} \text{Vol}\left(B_{\frac{1}{n}}(y_i^{(n)})\right) \leq s \text{Vol}\left(B_{\frac{1}{n}}(y_1^{(1)})\right) \xrightarrow{n \rightarrow \infty} 0,$$

Figure 1: The open-question sentence after Theorem 3.20 in arXiv:2503.24092.

Theorem 3.20 assumes that  $\Omega$  is a compact metric space without isolated points, that  $\mathcal{C}(\Omega, \mathbb{K})$  has a Schauder basis  $(b_j)_{j \geq 1}$  with coefficient functionals  $(c_j)_{j \geq 1}$ , and that  $k(n)$  is unbounded. In the nested case, the conclusion is that there are  $f \in \mathcal{C}(\Omega, \mathbb{K})$  and  $N \in \mathbb{N}$  such that for every  $n \geq N$ ,

$$(f(y_1), \dots, f(y_{k(n)})) \neq (c_1(f), \dots, c_{k(n)}(f)).$$

## Result

The literal non-nested analogue is false if the only size hypothesis is that  $k(n)$  is unbounded. The natural repaired statement is true if  $k(n) \rightarrow \infty$ .

**Proposition 1** (Counterexample to the literal unbounded formulation). *There are a compact metric space  $\Omega$  without isolated points, a Schauder basis of  $\mathcal{C}(\Omega, \mathbb{K})$ , an unbounded sequence  $k(n)$ , and non-nested sampling sets  $\{y_1^{(n)}, \dots, y_{k(n)}^{(n)}\}$  such that for every  $N$  and every  $f \in \mathcal{C}(\Omega, \mathbb{K})$  there is some  $n \geq N$  for which*

$$(f(y_1^{(n)}), \dots, f(y_{k(n)}^{(n)})) = (c_1(f), \dots, c_{k(n)}(f)).$$

*Proof.* Take  $\Omega = [0, 1]$ . Use the classical Faber–Schauder basis of  $\mathcal{C}([0, 1], \mathbb{K})$ , ordered so that its first coefficient functional is evaluation at the left endpoint:

$$c_1(f) = f(0).$$

This is the standard ordering of the Faber–Schauder system: the first basis vector is the constant function and the remaining basis vectors vanish at the left endpoint, so the first coordinate is  $f(0)$ .

Define

$$k(n) = \begin{cases} 1, & n \text{ odd}, \\ n, & n \text{ even}. \end{cases}$$

This sequence is unbounded. For every odd  $n$ , choose the one sampling point  $y_1^{(n)} = 0$ . The remaining, even-indexed sampling sets may be chosen arbitrarily.

For every odd  $n$  and every  $f \in \mathcal{C}([0, 1], \mathbb{K})$ ,

$$(f(y_1^{(n)})) = (f(0)) = (c_1(f)).$$

Thus the sampling encoder and the first basis encoder agree at infinitely many arbitrarily large indices. Consequently no single function  $f$  can make the two encoders differ for all sufficiently large  $n$ .  $\square$

**Theorem 1** (Repaired non-nested theorem). *Let  $\Omega$  be a compact metric space without isolated points and let  $(b_j)_{j \geq 1}$  be a Schauder basis of  $\mathcal{C}(\Omega, \mathbb{K})$  with coefficient functionals  $(c_j)_{j \geq 1}$ . Suppose  $k(n) \rightarrow \infty$  and for each  $n$  choose arbitrary sampling points*

$$y_1^{(n)}, \dots, y_{k(n)}^{(n)} \in \Omega.$$

*Then there are  $f \in \mathcal{C}(\Omega, \mathbb{K})$  and  $N \in \mathbb{N}$  such that for every  $n \geq N$ ,*

$$(f(y_1^{(n)}), \dots, f(y_{k(n)}^{(n)})) \neq (c_1(f), \dots, c_{k(n)}(f)).$$

*Proof.* For every  $n$ , define the continuous linear map

$$L_n : \mathcal{C}(\Omega, \mathbb{K}) \rightarrow \mathbb{K}^{k(n)}, \quad L_n(f) = (f(y_1^{(n)}), \dots, f(y_{k(n)}^{(n)})) - (c_1(f), \dots, c_{k(n)}(f)).$$

We first show that  $L_n$  is nonzero for all sufficiently large  $n$ . Suppose otherwise. Then there is an infinite set  $M \subseteq \mathbb{N}$  such that  $L_n = 0$  for every  $n \in M$ . Since  $k(n) \rightarrow \infty$ , for every fixed  $j$  there is  $n \in M$  with  $k(n) \geq j$ , and hence

$$c_j(f) = f(y_j^{(n)}) \quad \forall f \in \mathcal{C}(\Omega, \mathbb{K}).$$

If  $m, n \in M$  both satisfy  $k(m), k(n) \geq j$ , then evaluation at  $y_j^{(m)}$  and evaluation at  $y_j^{(n)}$  are the same functional. Continuous functions on the compact metric space  $\Omega$  separate points, so  $y_j^{(m)} = y_j^{(n)}$ . Therefore for each  $j$  there is a well-defined point  $y_j \in \Omega$  such that

$$c_j(f) = f(y_j) \quad \forall f \in \mathcal{C}(\Omega, \mathbb{K}).$$

It follows that every  $f \in \mathcal{C}(\Omega, \mathbb{K})$  has the Schauder expansion

$$f = \sum_{j=1}^{\infty} c_j(f) b_j = \sum_{j=1}^{\infty} f(y_j) b_j.$$

This contradicts Theorem B.2 of the source paper, which says that for a metric space without isolated points no Schauder basis of  $\mathcal{C}_b(\Omega, \mathbb{K})$  can have all coefficients given by sampling at a sequence of points. Since  $\Omega$  is compact,  $\mathcal{C}_b(\Omega, \mathbb{K}) = \mathcal{C}(\Omega, \mathbb{K})$ .

Thus there is  $N_0$  such that  $L_n \neq 0$  for all  $n \geq N_0$ . For such  $n$ ,  $\ker L_n$  is a proper closed linear subspace of the Banach space  $\mathcal{C}(\Omega, \mathbb{K})$ , and therefore it has empty interior. By the Baire category theorem,

$$\mathcal{C}(\Omega, \mathbb{K}) \neq \bigcup_{n \geq N_0} \ker L_n.$$

Choose  $f$  outside this union. Then  $L_n(f) \neq 0$  for every  $n \geq N_0$ , which is exactly the asserted eventual separation.  $\square$

## Limitations and interpretation

Proposition 1 refutes the literal replacement of nested sampling points by arbitrary non-nested sampling points under only the “unbounded  $k(n)$ ” hypothesis of Theorem 3.20.

Theorem 1 shows that the intended conclusion is recovered under the natural strengthening  $k(n) \rightarrow \infty$ . In particular, it applies to the usual sampling-encoder regime in which the sampling sets are  $1/n$ -coverings of an infinite compact metric space, since such covering cardinalities tend to infinity.

## Novelty check

The run indexes were searched for arXiv:2503.24092, the paper title, sampling encoders, basis encoders, universal operator approximation, and encoder-decoder keywords. No prior packet for this question was found. Bounded web searches for exact phrases from the open question, including “individual sampling points”, “sampling encoders”, and “basis encoders”, found the source arXiv page but no later answer.

## References

- [1] J. Goedeke and P. Fernsel, *New universal operator approximation theorem for encoder-decoder architectures (Preprint)*, arXiv:2503.24092, 2025.
- [2] Z. Semadeni, *Schauder Bases in Banach Spaces of Continuous Functions*, Springer, 1982.